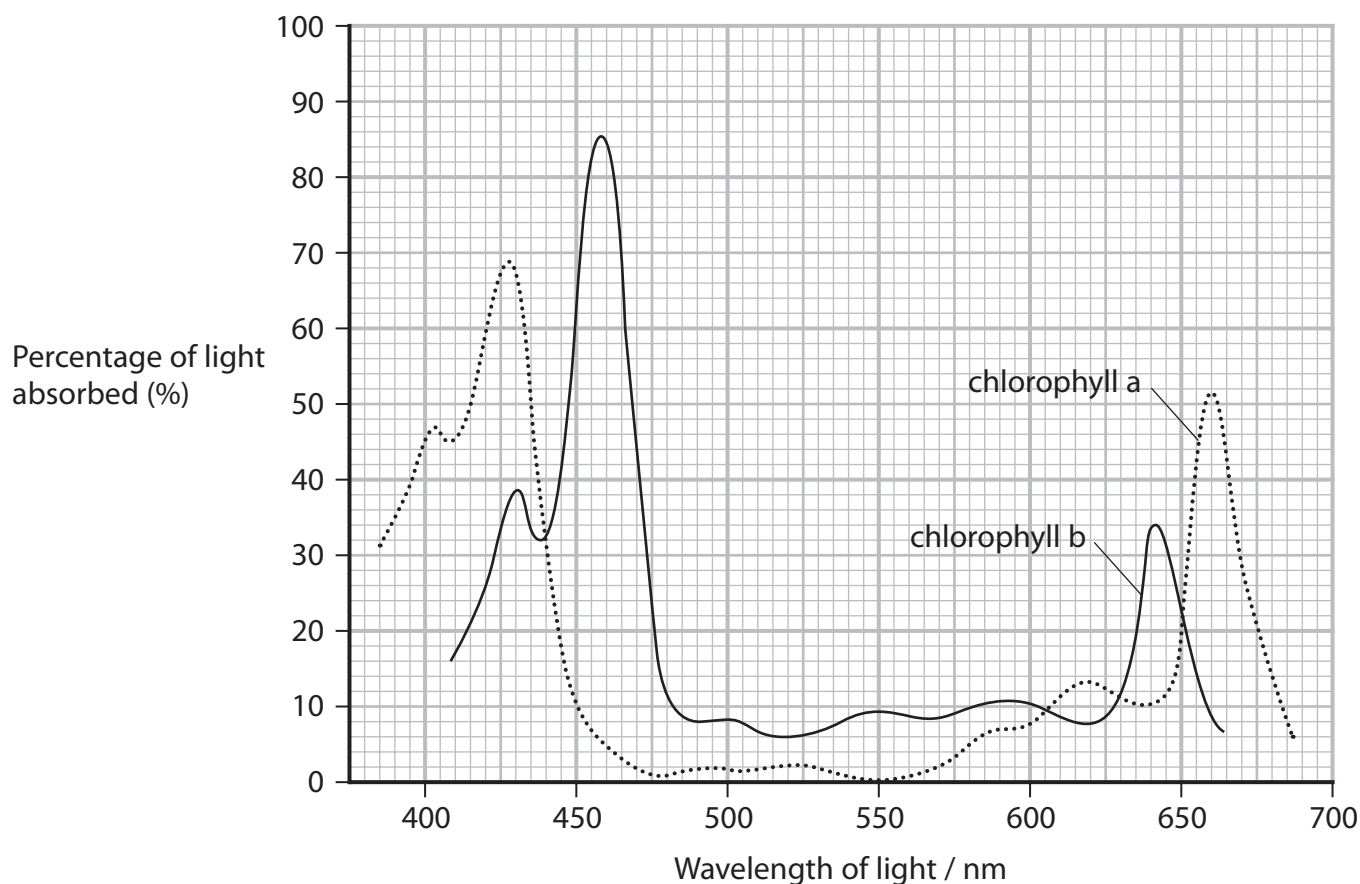


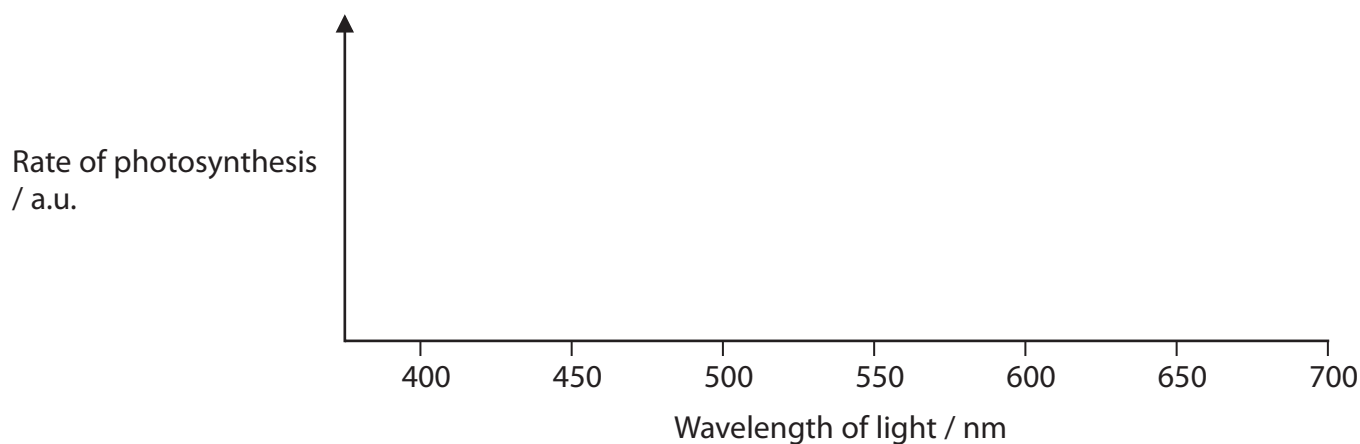
2 Thylakoid membranes contain chloroplast pigments and molecules of ATP synthase. These membranes are the site of the light-dependent reactions.

(a) The graph shows the absorption spectrums of two chloroplast pigments from one plant.



(i) Draw the action spectrum for this plant on the axes below.

(2)



- (ii) Calculate how many times greater the percentage of light absorbed by chlorophyll b is than that absorbed by chlorophyll a, at their maximum absorptions.

(1)

Answer

- (iii) Thylakoid membranes contain different types of chlorophyll pigment.

Explain the advantage of this for a plant.

(2)

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- (b) The ATP synthase molecules are proteins.

- (i) Which row of the table describes the bonds that form between amino acids in a protein?

(1)

	Name of bond	Parts of the amino acid that the bond forms between
☐ A	peptide	amino group and carboxyl group
☐ B	peptide	carboxyl group and R group
☐ C	phosphodiester	amino group and carboxyl group
☐ D	phosphodiester	carboxyl group and R group

- (ii) Products of the light-independent reactions are used to synthesise amino acids.

Describe how plants synthesise amino acids.

(2)

(Total for Question 2 = 8 marks)
